

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. CHB-503: Organic Chemistry-III

Time: 3 Hours

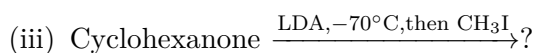
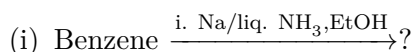
Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: $[5 \times 2 = 10]$

- (a) Why does benzophenone in isopropanol undergo photoreduction to pinacol upon irradiation at 345 nm?
- (b) What happens when 2-pentanone is irradiated with UV light? (Norrish Type I and Type II reactions).
- (c) What is the product when succinaldehyde is treated with ammonia?
- (d) What happens when pyridine is heated with KOH at 320°C?
- (e) What happens when an unsubstituted amide is treated with sodium hypobromite?

Question 2.

- (a) What is resolution of racemic modification? Explain with reference to the chemical method using chiral amine enantiomers. $[8]$
- (b) Predict the products of the following reactions: $[6]$

**Question 3.**

- (a) Outline the mechanisms of the following reactions: $[3 \times 4 = 12]$

(i) Pinacol-pinacolone rearrangement

(ii) Michaelis-Arbuzov reaction

(iii) Reformatsky reaction

- (b) Give one method for the synthesis of cyclobutane using malonic ester and 1, 3-dibromopropane. $[2]$

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